Logarithm: Formula Sheet

1. Exponential and log form

Exponential form: $N = a^x$

Log form: $\log_a N = x \iff N = a^x$

Conditions: $N > 0,\ a > 0,\ a \neq 1$

Same base: $a^x = a^y \Rightarrow x = y$

2. Basic values

Number = base: $\log_a a = 1$

Reciprocal:

$$\log_a \frac{1}{a} = -1$$

Number 1: $\log_a 1 = 0$

Power: $\log_a(a^p) = p$

Fundamental identity: $a^{\log_a N} = N$

3. Properties

Sum: $\log_a m + \log_a n = \log_a(mn)$

Difference:

$$\log_a m - \log_a n = \log_a \frac{m}{n}$$

Power of number: $\log_a(m^p) = p \log_a m$

Power of base:

$$\log_{a^p} m = \frac{1}{p} \log_a m$$

Reciprocal base: $\log_{1/a} m = -\log_a m$

4. Base change

Change of base:

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Chain: $\log_c a = (\log_b a)(\log_c b)$

Reciprocal:

$$\log_a b = \frac{1}{\log_b a}$$

Swap: $a^{\log_c b} = b^{\log_c a}$

Positive log: number and base on the same side of 1

Negative log: number and base on different sides of 1

5. Antilog, digits, zeroes

Antilog: $\text{antilog}_a(x) = a^x$

Characteristic and mantissa: $\log_{10} N = c + m,\ 0 \leq m < 1$

Scientific form: $N = p \times 10^n,\ 1 \leq p < 10$

Number of digits: $c + 1$

Zeroes after decimal: $|c + 1|$

6. Log equations

Remove log: $\log_a m = p \Rightarrow m = a^p$

Equal logs: $\log_a m = \log_a n \Rightarrow m = n$

Substitution: $t = \log_a x$ or $t = a^x\ (t > 0)$

Variable power: $f^g = h \Rightarrow g \log f = \log h$

Special case: $a^b = a^c \Rightarrow a = 1$ or $b = c$

7. Log and exponential inequalities

$a > 1$: $\log_a M > \log_a N \iff M > N > 0$

$0 < a < 1$: $\log_a M > \log_a N \iff 0 < M < N$

$a > 1$: $\log_a M > p \iff M > a^p$

$0 < a < 1$: $\log_a M > p \iff 0 < M < a^p$

$a > 1$: $a^x > a^y \iff x > y$

$0 < a < 1$: $a^x > a^y \iff x < y$

8. Modulus

Definition: $|x| = x\ (x \geq 0),\ -x\ (x < 0)$

Square: $|x|^2 = x^2, \quad \sqrt{x^2} = |x|$

Product: $|xy| = |x||y|$

$$|x + y| = |x| + |y| \iff xy \geq 0$$

$$|x - y| = |x| + |y| \iff xy \leq 0$$

9. Modulus equations and inequalities

Equation:

$$|f| = k\ (k \geq 0) \Rightarrow f = \pm k$$

Two moduli: $|f| = |g| \Rightarrow f^2 = g^2$

Less than: $|f| < k \iff -k < f < k$

Greater than: $|f| > k \iff f > k$ or $f < -k$

Two moduli: $|f| < |g| \iff (f - g)(f + g) < 0$

Negative k: $|f| < k$ none; $|f| > k$ all x in domain

10. Irrational inequalities

$$\sqrt{f} < g \iff f \geq 0,\ g > 0,\ f < g^2$$

$\sqrt{f} > g \iff [f \geq 0,\ g < 0]$ or $[g \geq 0,\ f > g^2]$

Nested radical: $\sqrt{a \pm 2\sqrt{b}} = \sqrt{p} \pm \sqrt{q},\ p + q = a,\ pq = b$